

Incident Management Automation

Client Scripting, UI Policies, and Automated Rules Validation Document

DEVELOPER / CONFIGURATOR	TARGET APPLICATION	INSTANCE ENVIRONMENT
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1. PROJECT OVERVIEW & PURPOSE

Incident records require consistent and accurate data entry to ensure effective triage, routing, and resolution. Relying solely on manual checks often leads to incomplete or incorrect data. The primary purpose of this project is to implement client-side controls using **UI Policies** and **Client Scripts** to enforce data integrity, automate field updates, and restrict unauthorized modifications directly on the Incident form and list views.

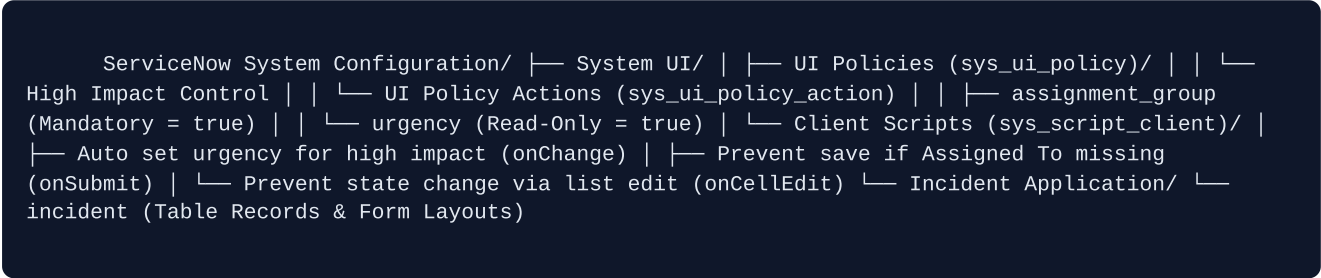
Dynamic UI Controls Automatically enforces read-only or mandatory field rules (e.g., locking Urgency and requiring Assignment Group for High Impact incidents).	Automated Field Logic Dynamically synchronizes fields by setting Urgency to "High" whenever Impact is set to "1 - High".
Submission Validation Intercepts form saving via client script validation, preventing save actions if mandatory assignments are missing.	List Editing Restrictions Prevents direct modifications to sensitive state fields directly from the list view without opening full audited forms.

2. ARCHITECTURAL OVERVIEW

The solution operates across the ServiceNow platform layer to ensure performant, responsive client execution without unnecessary server overhead:

- Client Layer (Browser):** Leverages the GlideForm (g_form) and GlideUser APIs to execute real-time logic immediately upon user field modification or form submission.
- Application Layer:** ServiceNow core Incident platform housing underlying Data Dictionaries, business rules, and UI definition tables (sys_ui_policy, sys_script_client).
- Database Layer:** Relational engine managing persistent records on the central incident table.

3. APPLICATION STRUCTURE & HIERARCHY

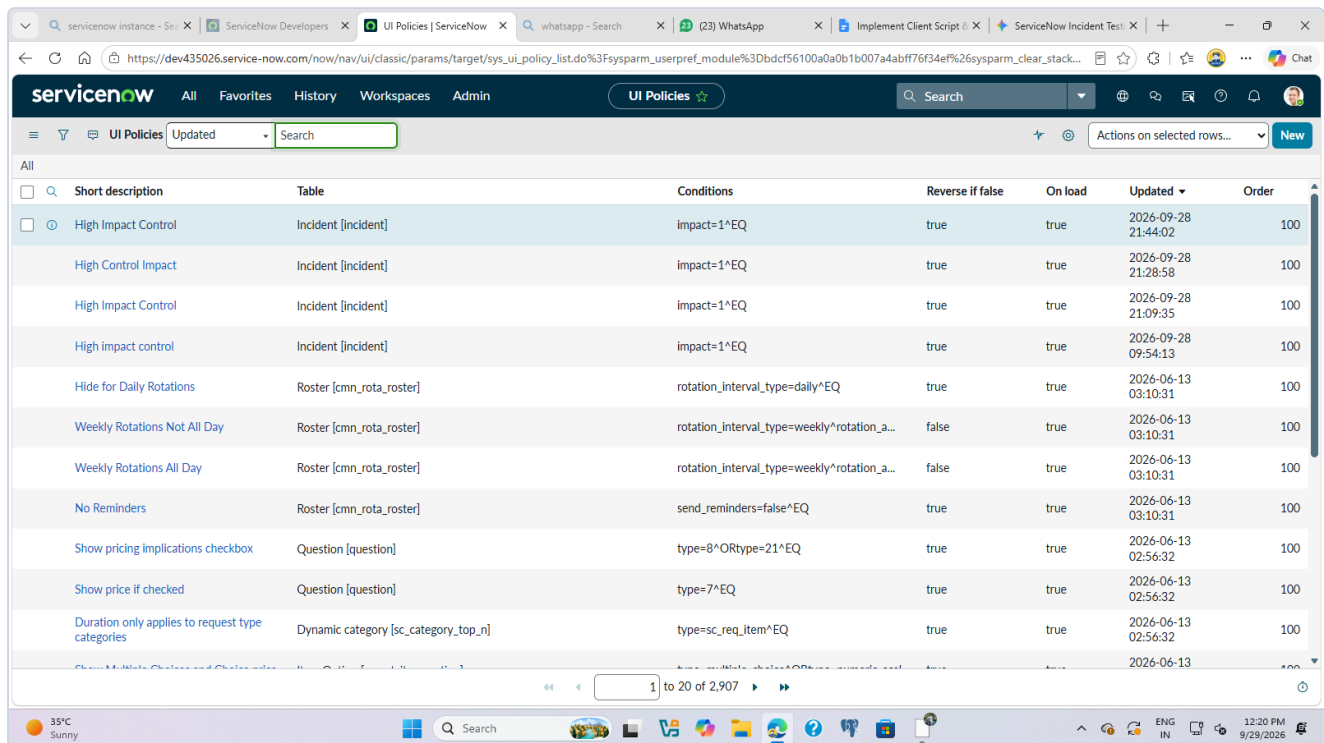


4. TECHNICAL SPECIFICATIONS & API MATRIX

The table below summarizes the client-side components and script logic configured to satisfy system requirements:

Name	Type / Event	Target Field	Trigger Condition	Function & Behavior
High Impact Control	UI Policy	urgency, assignment_group	Impact == 1 (High)	Sets Urgency to Read-Only and Assignment Group to Mandatory . Reverts when false.
Auto set urgency for high impact	onChange	impact	Field change (newValue == '1')	Executes <code>g_form.setValue('urgency', '1')</code> and displays systemic alert message.
Prevent save if Assigned To missing	onSubmit	Form Submission	Form Save (Impact == '1')	Evaluates if assigned_to is empty. Shows error pop-up and cancels submit by returning false.
Prevent state change via list edit	onCellEdit	state	Inline list edit action	Prompts confirmation dialog and calls <code>callback(false)</code> to reject list edit updates.

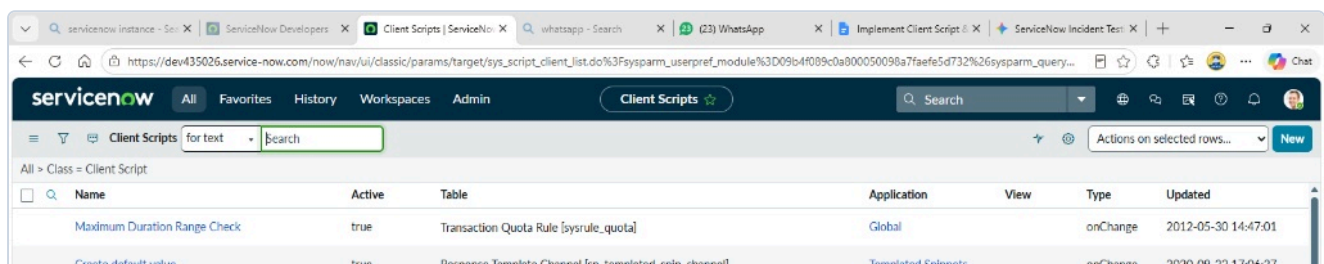
5. SYSTEM CONFIGURATION PROOFS (SCREENSHOTS)



The screenshot displays the ServiceNow 'UI Policies' configuration page. The table lists various policies, with 'High Impact Control' being the primary focus. It shows the target field 'urgency' and 'assignment_group', the trigger condition 'Impact == 1', and the function to set urgency to 'Read-Only' and assignment group to 'Mandatory'.

Short description	Table	Conditions	Reverse if false	On load	Updated	Order
High Impact Control	Incident [incident]	impact=1*EQ	true	true	2026-09-28 21:44:02	100
High Control Impact	Incident [incident]	impact=1*EQ	true	true	2026-09-28 21:28:58	100
High Impact Control	Incident [incident]	impact=1*EQ	true	true	2026-09-28 21:09:35	100
High impact control	Incident [incident]	impact=1*EQ	true	true	2026-09-28 09:54:13	100
Hide for Daily Rotations	Roster [cmn_rota_roster]	rotation_interval_type=daily*EQ	true	true	2026-06-13 03:10:31	100
Weekly Rotations Not All Day	Roster [cmn_rota_roster]	rotation_interval_type=weekly*rotation_a...	false	true	2026-06-13 03:10:31	100
Weekly Rotations All Day	Roster [cmn_rota_roster]	rotation_interval_type=weekly*rotation_a...	false	true	2026-06-13 03:10:31	100
No Reminders	Roster [cmn_rota_roster]	send_reminders=false*EQ	true	true	2026-06-13 03:10:31	100
Show pricing implications checkbox	Question [question]	type=8*ORtype=21*EQ	true	true	2026-06-13 02:56:32	100
Show price if checked	Question [question]	type=7*EQ	true	true	2026-06-13 02:56:32	100
Duration only applies to request type categories	Dynamic category [sc_category_top_n]	type=sc_req_item*EQ	true	true	2026-06-13 02:56:32	100

Figure 1: UI Policies active configuration list (sys_ui_policy) showing the active High Impact Control rule.



The screenshot displays the ServiceNow 'Client Scripts' configuration page. The table lists various scripts, including 'Maximum Duration Range Check' and 'Create default value'.

Name	Active	Table	Application	View	Type	Updated
Maximum Duration Range Check	true	Transaction Quota Rule [sysrule_quota]	Global		onChange	2012-05-30 14:47:01
Create default value	true	Response Template Channel [on templated snln_channel]	Templated Snlnnets		onChange	2020-09-22 17:06:27

6. USER INTERFACE EXECUTION & RUNTIME VERIFICATION

When an agent selects **1 - High** for Impact on an Incident record (e.g., INC0010006), the client scripts and UI policy execute immediately in sequence:

Incident - INC0010006

Urgency set to High for High impact incident.

Number: INC0010006

* Caller: Abel Tuter

Category: Inquiry / Help

Subcategory: -- None --

Service:

Service offering:

Configuration item:

* Short description: Testing high impact rules

Description:

Channel: -- None --

State: Resolved

Impact: 1 - High

Urgency: 1 - High

Priority: 1 - Critical

* Assignment group: CAB Approval

Assigned to: Bernard Laboy

Related Search Results >

Notes | Related Records | Resolution Information

Watch list:

Work notes list:

* Work notes:

Figure 3: Runtime execution on Incident INC0010006 – Urgency locked to Read-Only High, notification bar triggered, and fields updated.

Incidents | ServiceNow

INC0010006 has been resolved

All > Active = true

Number	Opened	Short description	Caller	Priority	State	Category	Assignment group	Assigned to	Updated	Updated by
INC0010006	2026-09-28 23:14:40	Testing high impact rules	Abel Tuter	3 - Moderate	Resolved	Inquiry / Help	CAB Approval	Bernard Laboy	2026-09-28 23:38:37	admin
INC0010005	2026-09-28 23:07:36	Hii	Abel Tuter	1 - Critical	In Progress	Inquiry / Help	CAB Approval	Bernard Laboy	2026-09-28 23:12:20	admin
INC0009009	2018-08-30 01:06:16	Unable to access the shared folder.	David Miller	4 - Low	New	Inquiry / Help	(empty)	(empty)	2018-12-12 23:30:24	admin
INC0009005	2018-08-31 21:35:21	Email server is down.	David Miller	1 - Critical	New	Software	(empty)	(empty)	2018-12-12 23:18:55	admin
INC0009001	2018-09-11 20:56:26	Unable to post content on a Wiki page	David Miller	3 - Moderate	New	Inquiry / Help	(empty)	(empty)	2018-12-12 23:32:42	admin
INC0008112	2019-07-29 11:48:43	Assessment : ATF Assessor	survey user	5 - Planning	New	Inquiry / Help	(empty)	(empty)	2019-07-29 11:49:28	admin
INC0008111	2019-07-22 14:04:57	ATF : Test1	System Administrator	5 - Planning	New	Inquiry / Help	(empty)	(empty)	2019-07-22 14:05:48	admin
INC0008001	2021-01-15 13:04:14	ATF:TEST2	survey user	5 - Planning	New	Inquiry / Help	(empty)	(empty)	2021-01-21 15:31:42	admin
INC0007002	2018-10-16 22:47:51	Need access to the common drive.	David Miller	4 - Low	New	Inquiry / Help	(empty)	(empty)	2018-12-12 23:28:49	admin
INC0007001	2018-10-16 22:47:10	Employee payroll application server is down.	David Miller	1 - Critical	New	Hardware	Openspace	(empty)	2026-06-13 03:47:11	system
INC0006999	2020-06-07	Unable to access the personal details	Problem	5 - Planning	On Hold	Inquiry / Help	(empty)	(empty)	2020-06-07	admin

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Figure 4: Incident list view confirming updated state (Resolved), Priority level (1 - Critical), and active assignment details.

7. QUALITY ASSURANCE & TEST CASE RESULTS

Test ID	Test Scenario	Expected Behavior	Actual Output	Status
TC-001	Mandatory Enforcement	Leave Assigned To empty on High Impact submission.	Submission aborted; error message displayed on form.	PASSED
TC-002	Successful Record Save	Populate Assigned To (Bernard Laboy) on High Impact incident.	Record saved successfully; Urgency locked to Read-Only.	PASSED
TC-003	UI Policy Reversal	Change Impact from High back to Medium.	Read-only lock on Urgency removed; fields return to editable state.	PASSED
TC-004	List Edit Restriction	Double-click State cell on Incident List view.	Alert modal displayed; inline list edit cancelled.	PASSED

8. SETUP & DEPLOYMENT STEPS

1. **Access Instance:** Log in to the ServiceNow PDI with admin credentials.
2. **UI Policy Creation:** Navigate to System UI → UI Policies and create the record on table incident with condition `impact=1^EQ`.
3. **Add Policy Actions:** Configure actions for urgency (Read-Only) and assignment_group (Mandatory).
4. **Client Scripts Setup:** Navigate to System UI → Client Scripts and register `onChange`, `onSubmit`, and `onCellEdit` handlers.

9. PROJECT METADATA & LINKS

- **GitHub Repository:** github.com/kokilamanogaran/ServiceNow-Incident-ClientScript-UIPolicy
- **Developer PDI:** <https://dev435026.service-now.com>

10. KNOWN ISSUES & FUTURE ENHANCEMENTS

- **Browser Cache Delays:** Client script modifications on `onCellEdit` may require force-refreshing browser cache during developer testing.
- **Server-Side Defense:** Implement parallel before `insert/update` Business Rules to prevent rule bypass via web service REST API imports.